CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of the first 11 multiples of 5 (that is 5, 10, ..., 55) is: 1

(a) 30 **(b)** 33 **(c)** 27 **(d)** 60

2. On a line graph, a town's population (in thousands) rises from 40 to 50. The percentage increase is: 1

(a) 10% **(b)** 35% **(c)** 12% **(d)** 25%

3. The mean mark of 12 students is 50 and of another 8 students is 75. The mean of all 20 students is: 1

(a) 60 **(b)** 62 **(c)** 62.5 **(d)** 1200

4. The mean of 21, 58, 54, 26, 41, 16 is 36. If 54 is replaced by 72, the new mean is: 1

(a) 36 **(b)** 39 **(c)** 54 **(d)** 48

5. The median of 86, 94, 97, 74, 20, 61, 72, 35 is: 1

(a) 73 **(b)** 80 **(c)** 67.38 **(d)** 20

6. The mean of 5 numbers is 26. The number to be added so that the mean becomes 33 is: 1

(a) 73 **(b)** 68 **(c)** 7 **(d)** 33

7. The mean of the first 15 multiples of 3 (that is 3, 6, ..., 45) is: 1

(a) 26 **(b)** 22 **(c)** 24 **(d)** 48

8. On a line graph, a town's population (in thousands) rises from 50 to 60. The percentage increase is: 1

(a) 20% **(b)** 30% **(c)** 10% **(d)** 22%

9. The mean mark of 25 students is 24 and of another 15 students is 40. The mean of all 40 students is: 1

(a) 33 **(b)** 1200 **(c)** 32 **(d)** 30

10. The mean of 27, 56, 55, 40, 19, 43 is 40. If 55 is replaced by 79, the new mean is: 1

(a) 44 **(b)** 40 **(c)** 64 **(d)** 53.17

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean of 5 observations 16, 13, 40, 11, 35 is 23. Find the mean if 9 is added to every observation. 2

12. Find the mean of the first 12 natural numbers that are multiples of 5. 2

13. The mean marks of 12 boys is 80 and of 18 girls is 70. Find the mean mark of the whole class of 30. 2

14. The mean of 5 observations 18, 10, 14, 30, 13 is 17. Find the mean if 12 is added to every observation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The mean weight of 8 students is 50 kg and of another 12 students is 45 kg. Find the mean weight of all of them. 3

16. The mean of 5 numbers is 19. If one number 19 is replaced by 29, find the new mean and the change in the mean. 3

17. The median of 76, 48, 63, 78, 25, 79, 42 is 63. If the largest value 79 is changed to 237, what happens to the median and the mean? 3

OR

The mean weight of 16 students is 38 kg and of another 24 students is 43 kg. Find the mean weight of all of them. 3

18. The mean of 5 numbers is 28. If one number 29 is replaced by 39, find the new mean and the change in the mean. 3

OR

The median of 77, 50, 43, 20, 10, 15, 75 is 43. If the largest value 77 is changed to 231, what happens to the median and the mean? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 15, 17, 24, 29, 28, 23. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 24 students has a mean mark of 45. Another 16 students with mean 60 join it. Find the mean of the combined group and the change in the original mean. 5

20. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 22, 26, 33, 36, 31, 25. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 30 students has a mean mark of 52. Another 20 students with mean 62 join it. Find the mean of the combined group and the change in the original mean. 5

END OF QUESTION PAPER — Total: 40 Marks